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Social inequalities in COVID-19-related years of life lost in metropolitan France, 2020

Romana Haneef^{1,2*}, Nour Mahrouseh³, Pascal Bessonneau¹, Stephanie Vandentorren^{4,5}, Nicolas Minier¹, Francis Chin⁶, Olivier Bruyère⁷, Brecht Devleeschauwer^{8,9} and Grant M. A. Wyper¹⁰

Abstract

Background The COVID-19 pandemic has directly, and indirectly, exacerbated existing social health inequalities. People living in socially deprived areas are at greater risk of most adverse outcomes from COVID-19. The aim of this study was to quantify the extent of social inequalities in COVID-19 related YLL (Years of Life Lost) in metropolitan France during 2020.

Methods The French national mortality database was used to identify COVID-19 related deaths. The French deprivation (FDep) index was used to assign deaths to a municipality-based social deprivation quintile of the French metropolitan population. Residual life expectancy at age of death was defined using the Global Burden of Disease (GBD) 2019 reference aspirational life table, and YLL was estimated for each registered death. Health inequalities were measured using gap measures between least and most deprived areas (absolute and relative), distributional measures (slope index of inequality (SII), relative index of inequality (RII)) and measure of potential impact (population attributable fraction (PAF)). The PAF was estimated using the least socially deprived quintile (Q1) as the reference level.

Results In 2020, the overall age standardized YLL rate related to COVID-19 per 100 000 population was 836, ranging from 831 in the least deprived quintile (Q1) to 1058 in the most deprived quintile (Q5), reflecting the non-linear distribution between deprivation quintiles and COVID-19 YLL. The absolute gap was 227 and relative range was 1.27. The SII was 234.3 [95% confidence interval (CI): 226.6–241.9] COVID-19 YLL age standardized per 100 000 population. The rate in the most deprived areas (Q5) was around 67% higher than the mean population rate (RII = 1.33). The PAF estimate was not consistent (PAF = 0.36%) with gap and distributional measures of inequality.

Conclusions COVID-19 cause-specific mortality was unequally socially distributed across areas. Our analysis highlighted marked inequalities in YLL estimates across urban and rural municipalities of France, the more disadvantaged a municipality, the higher YLL it has. Further research is needed to better investigate the mechanisms driving such inequalities, particularly regarding the wider role of social determinants of health, which has strong impact on the excess mortality risks.

Keywords COVID-19, YLL, ASYR, FDep, SII, RII, PAF, Health inequalities, Rural, Urban, France

*Correspondence:
Romana Haneef
Romana.HANEEF@santepubliquefrance.fr

Full list of author information is available at the end of the article



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Text box 1. Contributions to the literature

- There is limited evidence on the calculation of COVID-19-related YLL by levels of social deprivation using national data sources in France.
- This analysis contributes to highlighting marked inequalities in premature mortality associated with COVID-19 across urban and rural municipalities.
- A collaborative approach between policy makers and researchers is needed to address the underlying social determinants of health in order to improve the response to future public health crisis.

Background

The COVID-19 pandemic had an adverse population-level impact across all domains of life including health, but also economy, social interactions, and education [1, 2]. There is strong and clear evidence that COVID-19 pandemic has directly, and indirectly, exacerbated social health inequalities [2]. Health inequalities are expressed as the systematic, avoidable, and unfair differences in health outcomes that can be observed between populations, between social groups within the same population, or as a gradient across a population ranked by social position [3]. Due to the lack of exhaustive individual-level data on social health determinants, ecological measures (at a population level), such as deprivation indices where people live are usually used [4]. Various social deprivation indices have been developed in France to quantify the health inequalities [5, 6], relating to the concept set out by Townsend, « social disadvantage » or « deprivation », using the multiple aspects of poverty: income, employment, level of education, housing, etc [7]. There are several contextual and individual factors that are interdependent and play key roles in shaping health inequalities [8]. These factors can influence differential exposure to the virus, such as the social-residential environment including high population densities and overcrowded housing, over represented in social disadvantaged areas [9]. Ageing population is an important factor that can influence higher risk of mortality due to COVID-19 [10]. A study performed in France had showed that during the first lockdown (March 17 to May 10, 2020), a substantial decrease in hospitalizations for geriatric syndromes (GS) in older adults was accompanied by excess mortality [11]. Moreover, across countries, the proportion of older people (aged 75 years and above) in the population, gross domestic product per capita and unemployment rate were associated in 2020 with higher COVID-19 mortality rates [12]. Several comorbidities can also influence the differential global burden of diseases such as diabetes, hypertension, or overweight and obesity, which are most prevalent among people who reside in socially deprived areas [9]. Those people have a disproportionately greater risk of developing severe SARS-CoV-2 infection and dying [13, 14]. On the other hand, people living in more

socially advantaged areas are more likely to get tested for SARS-CoV-2, and less likely to test positive, be admitted to the hospital or intensive care, or die from COVID-19 than people living in more deprived areas [15].

In France, between January 10 and January 24, 2020 (the confirmation period of the first cases), nine possible cases were identified of which three cases were confirmed with COVID-19 [16]. Between February and the end of December 2020, two waves of the COVID-19 occurred in France: the first in March to mid-May 2020; and the second wave in October to November 2020 [17]. During the first wave, the circulation of virus was higher in the east than the south-west of France. To control the epidemic and keep the transmission rate low, the French government implemented two national lockdowns from March 17 to May 10, 2020 [18], and from October 28 to December 1, 2020 [19], with varying restrictions. An open-cohort study analyzing all-cause mortality in France found significant excess deaths during the COVID-19 pandemic, with 49,541 excess deaths (+ 8.0%) recorded in 2020 alone [20]. At the regional level, France experienced a reduction of nearly one full year in life expectancy in 2020 [21]. Antonio-Villa et al., used national death certificate data to examine socio-demographic inequalities and excess non-COVID-19 mortality during pandemic [22]. The direct impact of COVID-19 in France was substantial in 2020 in terms of disability-adjusted life years (DALYs), with almost all of health loss attributed to mortality (99%) [23].

To our knowledge, no study has investigated the social health inequalities in relation to years of life lost (YLL) associated with premature COVID-19 related mortality in France. There were two studies evaluating the impact of COVID-19 due to premature mortality attributable to social deprivation indices were performed, one in Scotland [4] and one in Chile [24]. We are aware that there were variations in COVID-19 response across countries due to several reasons such as political decisions to enforce lockdown restrictions, healthcare capacity, public attitudes, cultural factors, geography, existing social inequalities, etc. Therefore, we performed this study to investigate the impact of social health inequalities on the premature COVID-19 mortality in France. The aim of this study was to quantify the extent of social inequalities in COVID-19 related YLL in France during 2020.

Methods**Study population and period**

This is an ecological study setting in France, a country with a residential population of approximately 67 million in 2020. The assessment of COVID-19 YLL was conducted for the calendar year 2020. Social deprivation indices were defined using FDep Index for 2015 to analyse the geographical disparities in COVID-19

mortality at municipality level ($N=34\,874$, without overseas regions) [5]. This study is reported in adherence with the STROBE statement [25] (Supplementary material 1: Additional file 1).

French deprivation index

We used FDep index, a validated, multidimensional measure of socioeconomic status that captures income, education, employment, and housing data specific to the French context. It is widely used in public health research due to its relevance with health outcomes, including YLL, and its ability to identify regional disparities. The FDep index is an area-based proxy of social deprivation, and the score is constructed based on the 1st principal component (67–70% of the total variance explained depending on the period) of a population-weighted principal component analysis (PCA). It uses four aggregated variables: the percentage of unemployed individuals aged 15–64 years in the active population; the percentage of workers aged 15–64 years in the active population; the percentage of individuals aged 15 years and over with high-school certificates; and the median income per consumption unit [5]. The values of these four variables were derived from the 2015 census. This score is centered and reduced in order to assess the temporal evolutions. FDep scores are calculated only for metropolitan France by INSERM-CépiDc (*Institut national de la santé et de la recherche médicale -Center for Epidemiology on Medical Causes of deaths*) [26]. The index is not yet developed for the French overseas regions. In this study, we used the FDep as an indicator of deprivation of participants' place of residence on the municipality level. The first quintile (Q1) corresponds to the 20% of the population living in the least deprived municipalities, while the fifth quintile (Q5) corresponds to the 20% living in the most deprived municipalities. This represents approximately equal proportions of municipalities by quintile, without weighted by the number of inhabitants in each municipality. In 2015, INSERM-CépiDc updated the distribution of quintiles by weighting the number of inhabitants in each municipality and that corresponds to 12% municipalities with Q1, 21% to Q2, 22% to Q3, 25% to Q4 and 20% to Q5.

Mortality data source

In France, mortality data is treated by two main institutes: *first*, INSERM-CépiDc who is responsible to validate the medical causes of death by the trained experts [26]. *Second* is INSEE (*Institute of national statistics and economic studies*), which is responsible for the validation of civil status of the deceased person [27]. The validation of all related information is done in collaboration with INSEE and INSERM-CépiDc, using data from the medical section of the death certificate and civil status

certificate of death, to ensure the quality and standardization of all causes of death.

Individual-level data on death registrations were collected from the INSERM-CépiDc for the calendar year 2020 [26]. COVID-19 related deaths were defined as deaths reported with COVID-19 cause-specific as the main or underlying cause of death according to the World Health Organization (WHO) recommendations. COVID-19 related deaths were coded as having an ICD-10 code from U07.1 (virus identified) and U07.2 (virus not identified) as underlying cause of death, based on guidance from WHO [28]. We took into account deaths that were recorded with the municipality of residence listed. To integrate the area deprivation quintile, firstly, deaths at municipality level were extracted from INSERM-CépiDc mortality database, using INSEE-based codes. Then, these deaths were matched with the FDep index to assign the deprivation quintile at municipality level [6]. The number of deaths that did not match with FDep quintile were treated as missing data ($N=374$ (1% of the total deaths), all among overseas regions, i.e. 335 municipalities).

Up until 2020, INSEE defined “*rural*” as any municipality not part of an urban unit. An urban unit was characterized as a build-up area with more than 2000 inhabitants and a certain level of spatial continuity, typically associated with a “town”. This definition proposed here marks a break with the town-centered approach. Rural areas now denote all municipalities with a low or very low population density, according to a municipality density scale. In 2017, these municipalities represented 88% of all municipalities in France and accounted for 33% of the total population [29]. We accounted for the urban-rural dimension in the analysis to better understand the differences in YLL related to COVID-19.

Years of life lost (YLL)

YLL represents the total number of years that persons in a population would have lived if they had not died prematurely due to a disease or injury. It is calculated against an aspirational life expectancy, based on the lowest age-specific mortality rates observed across time and geographical locations [30]. The deaths at each specific age were weighted by the expected number of years lost at each age. The GBD 2019 reference life table was used for calculating the YLL for each age group [31]. We calculated age-standardised YLL rate (ASYRs) by applying the age structure of the 2013 European Standard Population [32], to the observed age-specific rates in France, which were calculated using French population. The age-standardized YLL rate is a key measure of premature mortality burden, allowing for comparisons across regions and socio-economic groups by adjusting for differences in age structure. It reflects not only how many people

died but also how early in life they died, which is critical for understanding the unequal impact of COVID-19 on population health. The ASYRs were presented based on five-year age groups and FDep quintiles. We also calculated the natural log transformation of these rates, which is used to normalize the distribution of YLL rates (which are often right-skewed), stabilize variance, and improve model fit in regression analyses. This transformation also allows for interpreting associations in relative terms, for example, the percent change in YLL associated with a unit increase in deprivation.

Measures of health inequalities

We calculated health inequalities to COVID-19 YLL by measuring the following indicators: (i) absolute and relative gap differences in YLL rates between the most (Q5) and least (Q1) deprived quintiles; and (ii) two distributional measures, namely relative index of inequality (RII) and slope index of inequality (SII), which account for the full distribution of deprivation across areas. These measures describe and quantify disparities in YLL across the deprivation gradient, providing insight into the extent of health inequality without implying a direct causal relationship. FDep was used to assess contribution of social deprivation to the outcome (i.e., COVID-19-related YLL). The SII provides an absolute measure of inequality while the RII provides a relative measure. To estimate the SII and RII, we followed the approach proposed by Moreno-Betancur et al. [33]. For RII, we fitted a Quasi-Poisson regression model, where each row contained age group, quintile of deprivation index, and the outcome, which is the counts of YLL with the population as an offset of that age group and quintile of deprivation index. The model includes fixed effects for age categories and midpoint. The exponential coefficient for midpoint represents the ratio between the bottom and top of the hierarchy of deprivation index. To estimate the age-standardized SII based on age groups, we followed the same approach by Moreno-Betancur et al. [33], using an additive Quasi-Poisson model (assuming a non-linear distribution). We obtained the SII for age category and then obtained a weighted sum of these SIIs (overall population), using weights from the 2013 European Standard Population. For both the RII and SII calculations, we used five-year age groups, pooling ages below 25 and defining the upper opened-ended group as 95 years and above. We estimated the population attributable fraction (PAF) of COVID-19-related YLL for each quintile using the following formula: $PAF = (V_{group}/V_{least} - 1) * (P_{group}/P_{all}) * 100$ [final result is a percentage], where, V_{group} = value for the group (numerator, rate, percentage), V_{least} = value for the least disadvantaged population, P_{all} = total population, P_{group} = population for the least disadvantaged population.

We performed a negative binomial regression analysis to assess the association between social deprivation quintile (i.e., FDep) and YLL related to COVID-19 across regions. The model was adjusted by age groups and area type (i.e., urban and rural), and population was used as an offset to account for differences in population size. *These estimates were expressed as incident rate ratio (IRR).*

Scenario-Based sensitivity analysis

To assess the robustness of our findings to the choice of socioeconomic index, we conducted a sensitivity analysis comparing age-standardized YLL rates (ASYRs) calculated using two different deprivation measures: the FDep index (French Deprivation Index) and the French version of the European Deprivation Index (EDI). This allowed us to examine whether the observed patterns in health inequalities remained consistent when alternative deprivation measures were applied. The two indices (FDep and EDI) are based on different components. The EDI index is calculated based on a combination of the following ten census-based variables aggregated at the area level associated with the individual level of deprivation: (1) proportion of individuals of foreign nationality, (2) proportion of households without a car, (3) proportion of individuals employed as managers or intermediate professionals, (4) proportion of single-parent families, (5) proportion of households with at least two individuals, (6) proportion of non-owner occupied households, (7) proportion of unemployed individuals, (8) proportion of individuals without post-secondary school education, (9) proportion of overcrowded dwellings, and (10) proportion of non-married individuals [6]. Like FDep index, the first quintile (Q1) represents the least deprived areas and the fifth quintile (Q5) represents the most deprived areas for EDI index. Further details were provided in supplementary material 1: Additional file 2.

Ethical approval

The study was based on aggregated and anonymous data from national mandatory mortality database (INSERM-CépiDc). This study does not require the ethics approval and consent to participate. In accordance with French law, Santé publique France (SpF) has been granted specific access to this national mortality database in order to carry out its mission of health monitoring and health crisis response (article L. 1413–7 code de la santé publique).

Results

The overall ASYR related to COVID-19 was 836 (95% CI: 834–838) per 100 000 population in France in 2020 (Fig. 1). The ASYR in least deprived quintile (Q1) was 831 (95% CI: 826–836) while the rate in the most deprived quintile (Q5) was 1058 (95% CI: 1052–1063). We observed a non-linear distribution of ASYRs across

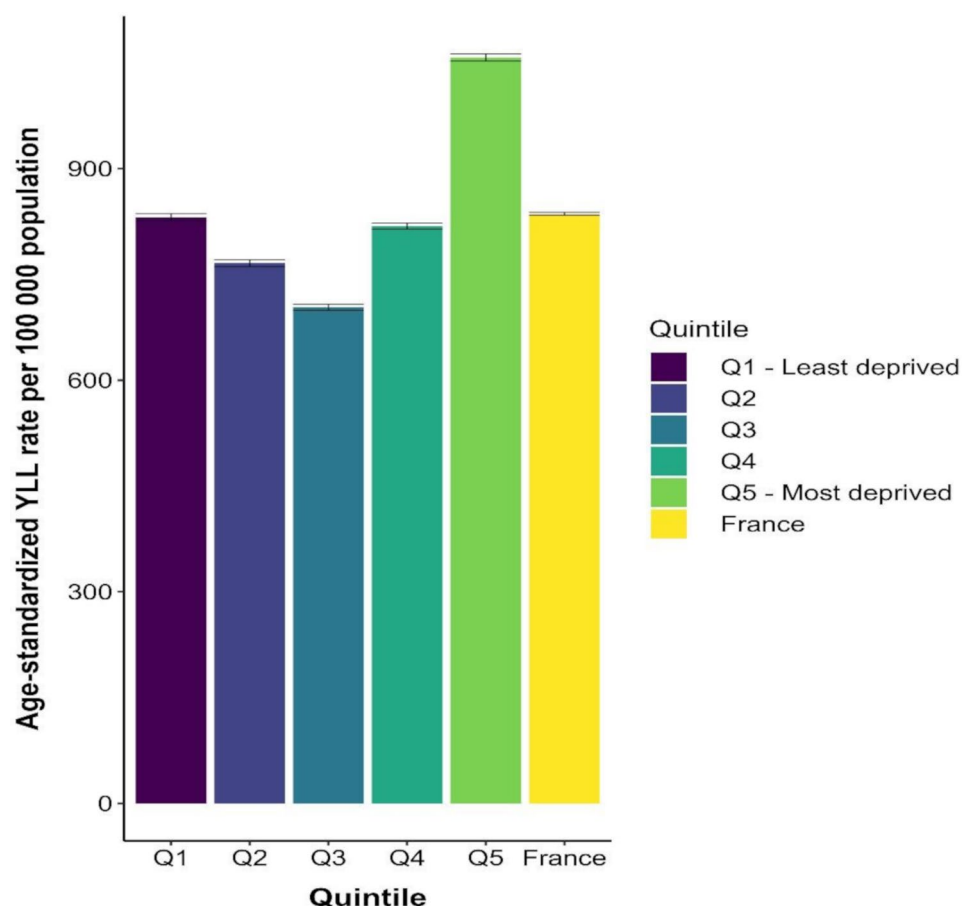


Fig. 1 Age standardized YLL rate related to COVID-19 per 100 000 population by FDep quintile, metropolitan France, 2020

quintiles, for example, Q3 had the lowest age-standardized YLL rate of 703 per 100 000 (95% CI: 699–708) (Fig. 1).

There were 34 874 municipalities (88% rural and 12% urban municipalities) with marked variations in COVID-19 mortality. Among the 30 711 rural municipalities, 26% had recorded at least one death related to COVID-19, whereas among the 4163 urban municipalities, 80% had recorded at least one death related to COVID-19. Q3 had the lowest age-standardized YLL rate related to COVID-19 and the variation of rural and urban municipalities was almost similar as above (i.e., 25% of rural municipalities had at least one COVID-19 death, whereas among urban municipalities 82% had recorded at least one death related to COVID-19).

We observed marked geographical variations in ASYR per 100 000 population at both rural and urban municipality levels. In rural municipalities, the ASYR rate ranged from 1,335 (Grand Est) in the least deprived quintile (Q1) to 3,519 (Centre-Val de Loire) in the most deprived quintile (Q5). On a natural logarithmic scale, this corresponds to an increase from $\ln(\text{ASYR})=7.19$ to 8.16. Exponentiating this change indicates that

COVID-19-related premature mortality was approximately 2.64 times higher in the most deprived rural areas, representing a 164% increase in YLL. This disparity underscores a pronounced socioeconomic gradient in the mortality burden within rural settings (Fig. 2). In contrast, urban municipalities exhibited an opposite trend. The ASYR rate declined from 643 (Normandy) in Q1 to 417 (Hauts-de-France) in Q5. On the logarithmic scale, this translates to a decrease from $\ln(\text{ASYR})=6.47$ to 6.03. Exponentiation reveals that COVID-19-related YLL was about 36% lower in the most deprived urban areas compared to the least deprived. This reverse gradient suggests that, unlike rural areas, deprivation in urban settings was not associated with higher mortality. Possible explanations may include differences in age distribution, exposure patterns, or institutional characteristics across urban municipalities (Fig. 2).

The age-standardized YLL rates were higher in the east, followed by southeast municipalities (Fig. 3a). There were many municipalities located in the west and southwest where no COVID-19 deaths were recorded (Fig. 3a). Figure 3b illustrates geographical variation of the FDep by municipality. The less deprived municipalities (i.e., with

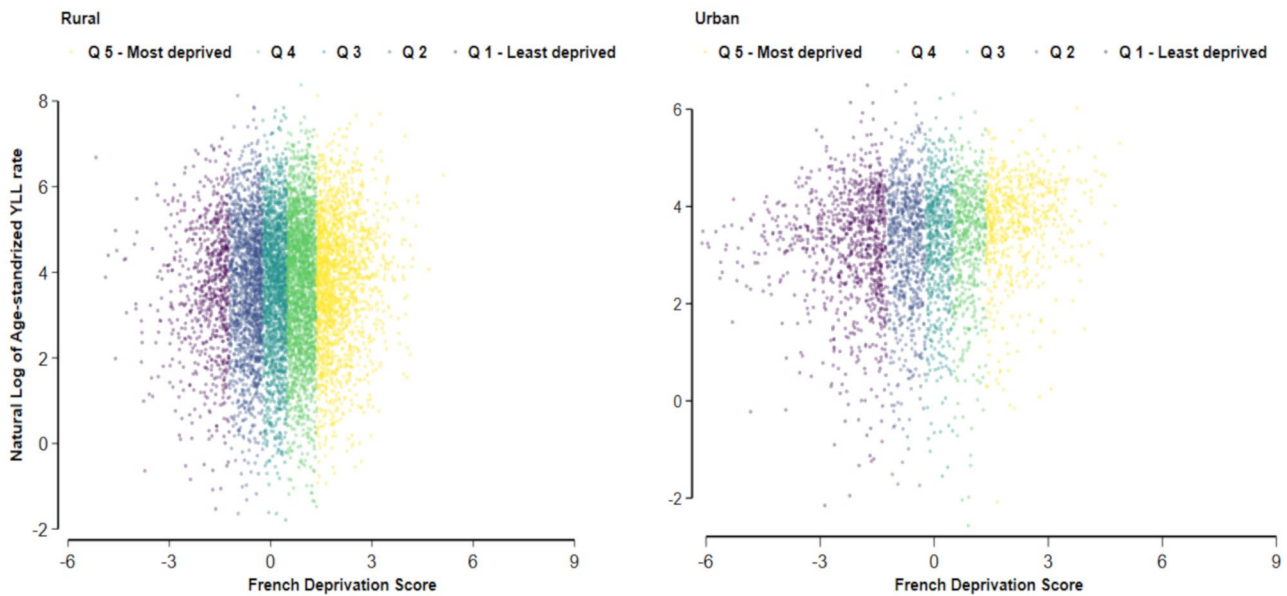


Fig. 2 Natural log of age standardized YLL rate per 100 000 population by quintile and French Deprivation score at municipality level, in metropolitan France, 2020

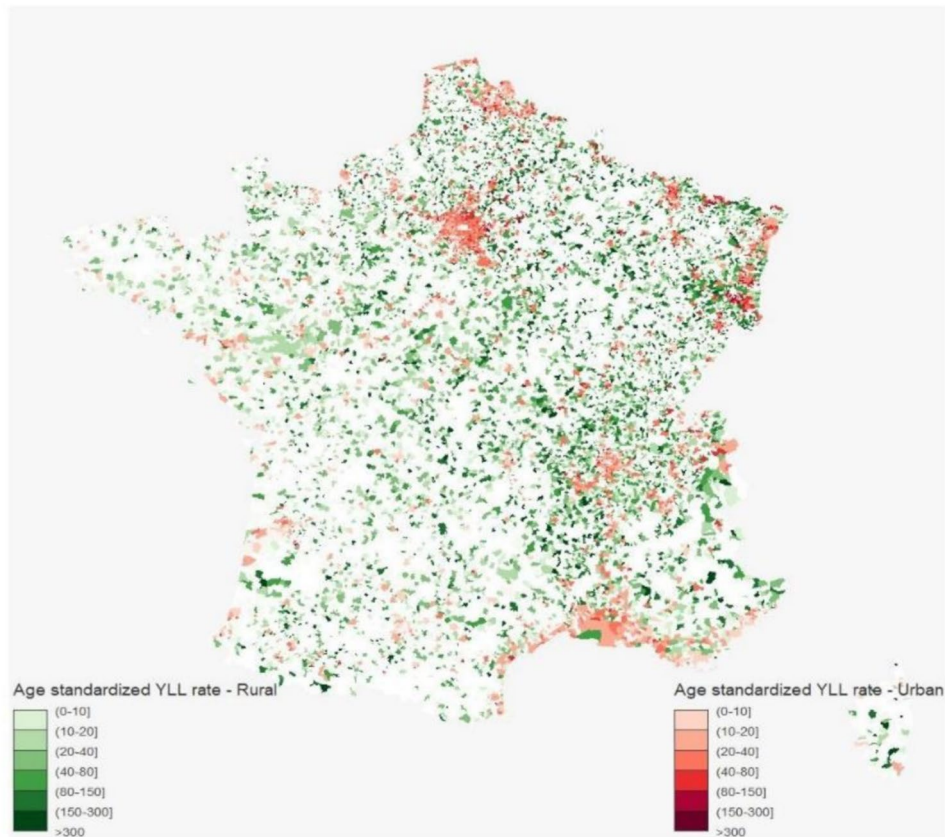


Fig. 3a Age standardized YLL rate per 100 000 population by rural and urban municipalities in metropolitan France, 2020

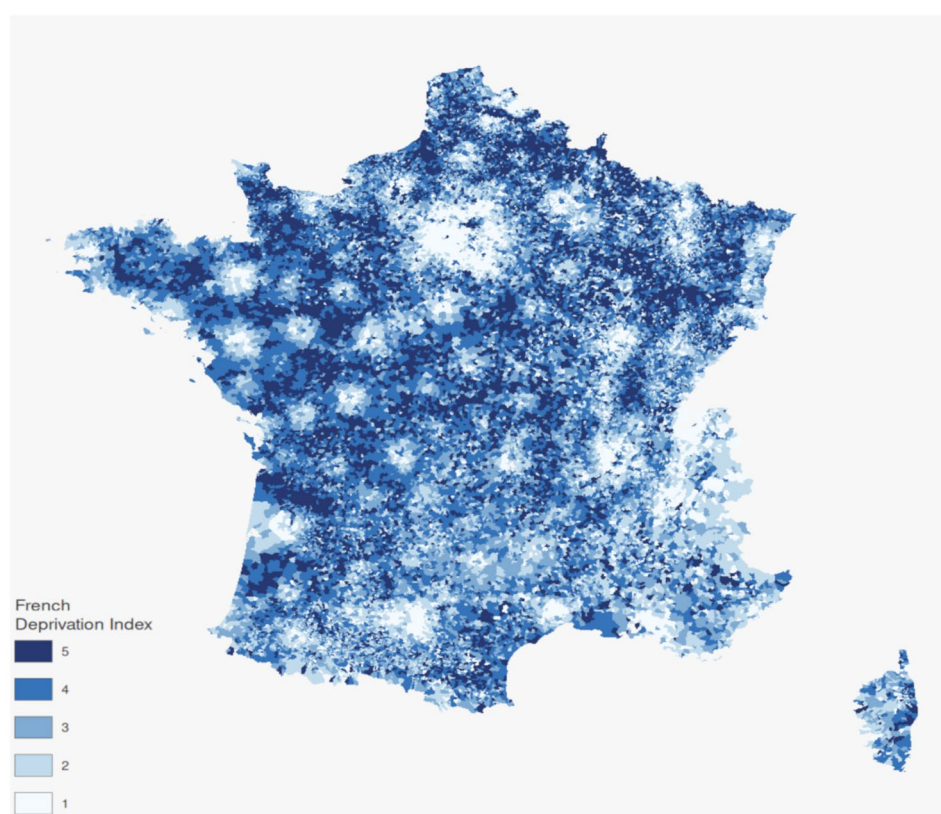


Fig. 3b French Deprivation Index by quintile and municipality in metropolitan France, 2015

Table 1 Health inequalities in age-standardized YLL rates related to COVID-19, in France, 2020

Measures of health inequality	YLL
Absolute range	226.67
Relative range	1.27
Relative index of inequality (RII) (based on age groups)	1.338: 95% CI [1.325–1.350]
Age-standardized Slope index of inequality (SII) (based on age groups) per 100,000	234.3: 95% CI [226.6–241.9]
Population Attributable Fraction (PAF)	0.36%

lowest deprivation level) were clustered around large cities, in the Paris area and near the Swiss border.

Measures of health inequalities

The absolute difference in ASYRs per 100 000 population was 227. The relative difference in ASYRs between the most (Q5) and least (Q1) deprived areas was 1.27, indicating that YLL was 27% higher in most deprived areas as compared to the least deprived areas (Table 1). The difference between the most and least deprived areas, measured by SII, was 38 199 (95% CI: 36 932–39 465) YLL rates. The age adjusted RII was 1.33, which when multiplied by 0.5 and expressed as a percentage indicates that the rate in most deprived areas was 67% higher than the mean population rate (i.e., 836 ASYR). The age

standardized SII was 234.3 (95% CI: 226.6–241.9) which indicates that even after adjusting for age, there remains a difference in YLL attributable to social deprivation between the most deprived areas and the least deprived ones. The PAF estimate contrasted with other measures. Indeed, we found that only 0.36% of COVID-19-related YLL rates at national level was explained by inequalities measured by FDep quintiles, at national level. However, these estimates varied by area: in rural areas, the most deprived population experienced lower YLL rates, while in urban areas, a modest proportion of COVID-19-related YLL rates was attributable to deprivation (Supplementary material 1: Additional file 3).

We also report these results separately by rural and urban areas in « Supplementary material 1 : Additional file 3 and file 4 ».

We observed regional variations in the association between FDep quintile and YLL related to COVID-19. Compared to the reference region Bretagne (with lowest YLL rate), most regions showed significantly higher YLL. Most notably, Ile-de-France with over 12 times higher YLL than the reference region, followed by Bourgogne-Franche-Comté with 6.2 times and Grand-Est 5.6 times increase. Pays de la Loire did not show significant association with YLL compared to the reference region. Each increase in deprivation quintile from 1 to 5 was

associated with a 17% higher YLL. The deprivation index was statistically significantly associated with substantial increased YLL, whereas the area type was not statistically significant (Supplementary material: Additional file 5). Figure 1 showed the U-shape distribution whereas the ASYR_EDI showed a linear distribution with the highest values observed in Q5 (most deprived) (Supplementary material 1: Additional file 6). In the scenario based sensitivity analysis, we compared the compared age-standardized YLL rates (ASYRs) using two different deprivation indices: the French Deprivation Index (FDep) and the French version of the European Deprivation Index (EDI). Further detail was provided in Supplementary material 1: Additional file 7.

Discussion

Our study found that COVID-19 cause-specific mortality was unequally distributed across areas with varying levels of social deprivation. These results showed a non-linear distribution of age-standardized YLL rates related to COVID-19 according to social deprivation index, nevertheless illustrating the stronger and disproportionate effect on Q5, highest among the most deprived areas. Across both rural and urban municipalities, we observed marked variations in YLLs rates, higher in the east, followed by south-east municipalities of France when compared with west and south-west municipalities. Overall, our results highlighted the absolute and relative inequalities in YLL estimates. The PAF estimates appears inconsistent with the more pronounced disparities observed through absolute and relative gap measures (Q5 vs. Q1) and the SII/RII metrics. The PAF assumes a binary comparison between the most deprived group and a counterfactual scenario where the entire population experiences the risk of the least deprived group (Q1). As such, it is sensitive to the size of the most deprived population group and does not account for the full socioeconomic gradient. In contrast, gap-based measures (e.g., Q5–Q1 differences) and distributional measures like the SII and RII capture the entire range of deprivation, providing a broader view of inequality across quintiles. Moreover, the relatively small PAF value may be partially explained by the overall distribution of YLL across quintiles, where even the least deprived groups still experienced substantial YLL during the pandemic. This dilutes the relative “gain” that could theoretically be achieved by shifting the entire population to the lowest-risk category. Moreover, when stratified by area type, socioeconomic inequalities in COVID-19 related mortality were more pronounced in urban than rural areas. Urban settings showed a clear gradient with higher YLL rates among the most disadvantaged, while rural areas exhibited minimal or reversed inequality patterns. These differences likely reflect variations in exposure risk and population density. Regional

disparities were also evident in YLL related to COVID-19, with most regions showing higher YLL compared to Bretagne with Ile-de-France having the highest increase. Deprivation index was associated with substantially increased YLL. These results were not consistent with findings from studies conducted in other countries, which concluded to a linear gradient across deprivation indices using mortality rates and DALYs [4, 34, 35]. Another study conducted in Chile showed an heterogeneous geographical distribution by social deprivation quintile status during the first wave (3 March to 10 October 2020) of COVID-19 [24]. There are some key factors, which could potentially contribute to the socially differentiated impact of COVID-19, and could therefore have an impact on these results. For example, the ageing, comorbidities, distribution of rural and urban municipalities with and without COVID-19 deaths and limited access to health care services. *First*, we observed differences in the age distribution of COVID-19 deaths across deprivation levels and geographic areas, although our mortality metrics are age-standardized. In urban municipalities, the least deprived quintiles had a greater number of people aged 70 and above who died from COVID-19 compared to the most deprived areas. In contrast, rural municipalities showed the opposite pattern: the most deprived areas had older individuals among the COVID-19 deaths. Overall, the average age of death from COVID-19 varied across regions, ranging from 80 to almost 85 years (Supplementary material 1: Additional file 8). These observations highlight the concentration of older, vulnerable populations in different contexts, which may influence health system needs and pandemic impact beyond what is captured by age-standardized rates. Despite these demographic differences, our results showed that YLL rates were consistently higher in the most deprived quintile (Q5), emphasizing the disproportionate burden among socioeconomically disadvantaged populations. A national cohort study in France emphasized that age is the main factor of developing severe forms of COVID-19 and a number of comorbidities (cardiovascular, obesity, hypertension, etc.) were associated with a higher risk of hospitalization and in-hospital death for covid-19 [36]. *Second*, we observed a marked difference in the distribution of COVID-19 deaths that were mainly recorded in urban municipalities as compared to the rural municipalities (i.e., 80% vs. 26%, respectively). This variation reflects that higher population living in urban municipalities, most importantly the differential circulation and transmission of COVID-19 virus among these municipalities, respecting various measures such as wearing mask, staying at home, etc., which could potentially contribute to the COVID-19 death, thus might influence a non-linear distribution of YLL. *Furthermore*, a study performed in France indicating that the

most socioeconomically disadvantaged municipalities were affected the most during the first year of COVID-19 mortality [37]. *Third*, health care access plays a crucial role. Evidence suggests that individuals living in most deprived areas may face barriers to accessing timely and high-quality health care services, such as COVID-19 testing in France [9]. The use of certain healthcare services seems to be suboptimal among the most deprived populations [38–40], potentially contributing to delayed diagnosis and inadequate treatment, ultimately leading to higher mortality rates among most deprived populations. Selective adjustment of cause-specific YLL estimates can divert attention from the environmental and risk factors driving mortality. YLL is a critical metric for informing public health policy. Comorbidity adjustments risk disadvantaging socioeconomically deprived populations with higher morbidity burdens, potentially deprioritizing them in resource-limited contexts and worsening health inequities. Such approaches should be avoided when aiming to identify and address unjust disparities in population health [41]. In sum, these factors likely contribute both directly or indirectly to the COVID-19 YLL rates and highlight the complex interplay between various biological, environmental, behavioural, and social factors, in shaping the unequal impact of the pandemic.

Strengths and limitations

This is the first study that quantifies the COVID-19-related YLL across municipalities in France in 2020, with a focus on disparities by social deprivation. We obtained data from INSERM-CépiDc, which relies on a robust vital registration system at the national level in France. We integrated the FDep index at the municipality level to each recorded COVID-19 death to capture the social deprivation at a finer geographical scale. There are some limitations. *First*, the absence of comparable pre-pandemic data at the municipal level, which prevents us from directly assessing whether the observed inequalities in COVID-19-related premature mortality represent a widening or continuation of existing health disparities. *Second*, we used the GBD reference life table to facilitate the comparison of these results with other studies, which is consistent with GBD and WHO methods. Nevertheless, we acknowledge that the choice between national and reference life tables is of intense debate [42]. *Third*, we did not measure the COVID-19-related YLD by social deprivation, as the majority of health loss was due to mortality (i.e., 99%) [23]. Therefore, we focussed on analyzing the extent of social deprivation to YLL estimates associated with COVID-19. *Fourth*, the FDep index only assesses four socioeconomic variables at area level, which may limit its ability to capture health inequalities contributed by several other aspects such as health, access to health care services, housing and crime at individual level.

Therefore, to better capture the inequalities, it is essential to investigate individual socioeconomic determinants in order to understand the underlying mechanisms driving these health disparities. *Fifth*, we did not yet calculate all-cause mortality according to the area deprivation levels, to compare the relative contribution of COVID-19 YLL to other causes of death. Further research will examine how social deprivation influences all-cause mortality and assess the relative burden of COVID-19-related deaths in comparison to other causes of death. *Finally*, the present study is limited in scope to the year of 2020, due to a discontinuation of the routine calculation of YLL. Nevertheless, we believe that the 2020 findings remain highly relevant, as they capture the critical initial phase of the pandemic when mortality disparities were most pronounced.

We observed a marked difference between two the measurements, ASYR_FDep and ASYR_EDJ, used in the scenario analysis. The choice between FDep and EDJ remains a subject of debate, as each index may yield differing assessments of socioeconomic disparities. Previous studies have discussed that results can vary depending on the deprivation indices used [43–45] and our findings align with this observation. As explained in previous studies, this difference can be attributed to several reasons: *First*, the EDJ tends to classify urban municipalities in more deprived quintiles, whereas the FDep tends to classify them in less deprived quintiles, particularly for Paris and suburbs [43, 44]. This can partly be explained by the fact that the FDep and the EDJ created in different contexts and have different objectives [44]. *Second*, these indices were built using different statistical methods [44]. The FDep is built from four ecological variables to capture and maximize the spatial variability through the statistical procedure of principal component analysis, and it has been validated using mortality data [5]. In contrast, the EDJ is built from a combination of ten weighted census-derived variables, chosen as a proxy of individual deprivation experience and average social deprivation [6], which can be less adapted to capture the variety of socio-spatial situation that composed the French territory [43]. *Third*, the FDep was developed in the context of ecological approaches, whereas the EDJ was constructed to proxy individual socioeconomic position [43]. As a result, the different components in the two deprivation indices capture ecological inequalities in different ways. Further research is needed to reach a consensus on which kind of area-based ecological deprivation indices should be used to assess the social inequalities of health in France by the researchers.

Implications in policy and further research

Our findings showed the impact of social inequality on COVID-19 YLL that has occurred across the whole

population of France, which is unequally distributed. These results have several important implications for policy and further research that can be extrapolated to other countries with similar social structures and health disparities: **Policy implications:** Our findings highlighted significant geographic and socioeconomic inequalities in premature mortality during the COVID-19 pandemic, particularly with disproportionately higher YLL rates in the most deprived areas. These results underline that it is essential to invest in health care infrastructure in socioeconomically deprived areas to ensure that quality health care services are accessible to all. In particular, strengthening access to primary care, preventive services, and tailored community-based interventions in high-burden areas could help mitigate avoidable mortality. Policymakers should interpret the PAF estimates jointly to appreciate both the burden concentrated in the most deprived groups and the broader pattern of social disparities across the population. Policy measures should also address the upstream social determinants that contribute to excess mortality, including income insecurity, substandard housing, and limited access to health information. For example, income support programs and housing improvements could reduce vulnerability among deprived populations. In addition, non-stigmatizing public health education campaigns should be expanded to promote healthy behaviors across all communities, particularly in areas where structural disadvantage overlaps with health risk. Given that France exhibits higher levels of mortality inequality than many European countries, and that these inequalities have widened in recent years reducing health disparities must be a policy priority informed by evidence such as the findings presented in this study [46]. **Further research implications:** Our results highlighted marked differences among the municipalities with and without COVID-19 death. Therefore, it appears important to compare these municipalities by taking into account health care access, behavioural risk factors, and social determinants of health such as income, housing, low-wage professions. Furthermore, the differences in YLL observed across quintiles are primarily driven by the absolute number of deaths, rather than by differences in age-specific remaining life expectancy. Applying deprivation-specific life expectancies could potentially reduce the observed YLL gap between the least and most deprived groups. Further research is warranted to incorporate life tables from the Human Mortality Database (HMD) or to better construct stratified life tables by deprivation quintile to better capture the impact of socioeconomic differences on YLL estimates. These approaches could enhance the precision and contextual relevance of YLL estimates within the national burden of disease framework. Further research is needed to better understand the mechanisms underlying these

health inequalities, particularly broader social determinants of health not captured by the FDep index, such as healthcare access, education, housing, and living conditions, which can affect mortality and YLL rates. Future studies should also examine the association between individual social determinants of health and YLL, while adjusting for potential confounders. Additionally, with COVID-19 vaccination programs introduced in France in 2021, further analysis is warranted to assess the impact of vaccination across socioeconomic gradients.

Conclusions

COVID-19 cause-specific mortality was socially unequally distributed across areas according to different levels of social deprivation. Our analysis highlighted marked variations in YLL estimates across municipalities of France. These results underline the critical need for targeted and equitable public health strategies to address the disproportionate impact of COVID-19 on socioeconomically disadvantaged populations. Further research is needed to better understand the mechanisms driving such health inequalities, especially broader social determinants of health, which can impact mortality. Specifically, future studies are essential to examine the association between individual social determinants of health and YLL, taking potential confounding factors into account. The implementation of effective policy interventions to reduce health inequalities should be the priority plan to improve population health and future pandemic responses should be considered of reducing health inequalities. Policymakers and researchers should collaborate to design, implement, and evaluate interventions that not only address immediate health needs but also tackle the underlying social determinants of health inequalities, in order to reduce overall health disparities and improve resilience against future public health crises.

Abbreviations

COVID-19	Coronavirus Disease 2019
YLL	Years of Life Lost
ASyr	Age-Standardised YLL rate
FDep	French Deprivation Index
EDI	European Deprivation Index
GBD	Global Burden of Disease
SII	Slope Index of Inequality
RII	Relative Index of Inequality
PAF	Population Attributable Fraction
Q	Quintile
GS	Geriatric Syndromes
DALYs	Disability Adjusted Life Years
INSERM	Institut national de la santé et de la recherche médicale
CépiDc	Centre d'épidémiologie sur les causes médicales de Décès de l'Inserm
INSEE	Institute national de la statistique et des études économiques
WHO	World Health Organization

Supplementary Information

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Supplementary material 1: **Additional file 1:** It is a doc. Word file. It reports the STROBOD checklist of items in report of Disability-adjusted life year calculations. **Additional file 2:** It is a doc. Word file. It describes the Scenario-Based Sensitivity Analysis. **Additional file 3:** It is a doc. Word file. It reports Age-standardized YLL rates (ASYRs) and health inequalities indicators related to COVID-19, by rural and urban areas in France, 2020. **Additional file 4:** It is a doc. Word file. It visualizes the Age standardized YLL rate (ASYR) related to COVID-19 per 100 000 population by FDep quintile, by rural and urban metropolitan France, 2020, as Fig. S1. **Additional file 5:** It is a doc. Word file. It reports the results of regression analysis between FDep quintile and YLL related to COVID-19 in 2020 across metropolitan regions in France, 2020. **Additional file 6:** It is a doc. Word file. It visualizes the Age standardized YLL rates related to COVID-19 per 100 000 population by EDI (European Deprivation Index) quintile, metropolitan France, 2020, as Fig. S2. **Additional file 7:** It is a doc. Word file. It describes the results of "Scenario-Based Sensitivity Analysis" and visualized the results, as Fig. S3. **Additional file 8:** It is a doc. Word file. It described the average age of COVID-19 death by regions in 2020

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Author contributions

RH, NM, GW generated the idea for the study. RH, NM, BD, GW developed the methodological approach. RH and FC supported data curation. RH and NM carried out all the analyses. RH, NM, PB and NM contributed to the visualization of results. RH drafted the original manuscript. SV, OB, BD, and G.M.A. Wyper provided critical input into the methodology and interpretation of the results. All authors reviewed and approved the final draft of the manuscript.

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Data availability

No datasets were generated or analysed during the current study.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

OB, BD and GW are co-authors of this manuscript; the former are the Editors-in-Chief, and the latter is the section editor of Archives of Public Health. The other authors declare no competing interests.

Author details

¹Department of Non-Communicable Diseases and Injuries, Santé publique France, Saint-Maurice, France

²CERPOP, UMR1295, unite mixte INSERM – University of Toulouse III Paul Sabatier, Toulouse, France

³Department of Public Health and Epidemiology, Faculty of Medicine, University of Debrecen, Debrecen, Hungary

⁴Departement of scientific direction and international, Santé publique France, Saint-Maurice, France

⁵Bordeaux Population Health Laboratory, INSERM U1219, University of Bordeaux, Bordeaux, France

⁶Department of Data science, Santé publique France, Saint-Maurice, France

⁷Research Unit in Public Health, Epidemiology and Health Economics, University of Liège, Liège, Belgium

⁸Department of Epidemiology and Public Health, Sciensano, Brussels, Belgium

⁹Department of Translational Physiology, Infectiology and Public Health, Ghent University, Merelbeke, Belgium

¹⁰School of Health & Wellbeing, University of Glasgow, Glasgow, UK

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